Magnetic Monopole: Wave Equation with Variable Coefficient in Cylindrical Coordinates 1. Problem Statement

We solve the wave equation with variable coefficient under axial cylindrical symmetry (f independent of azimuthal angle θ): $\frac{R}{r}\,\Delta f(r,z,t) = \frac{1}{c^2}\frac{\partial^2 f(r,z,t)}{\partial t^2}$

where the cylindrical Laplacian with axial symmetry ($\partial/\partial\theta = 0$) is:

$$\Delta f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{\partial^2 f}{\partial z^2} = \frac{\partial^2 f}{\partial r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{\partial^2 f}{\partial z^2}$$

Boundary / initial conditions: $f(2R,\,0,\,0) = 0, \qquad \frac{\partial f}{\partial r}|_{(0,0,0)} = -R$

with $R = 3.9228 \times 10^{22}$ m and $c = 2.998 \times 10^{8}$ m/s.

2. Separation of Variables Assume a product solution:

$$f(r,z,t) = H_3(r)\,H_4(z)\,u(t)$$

Substituting into the PDE: $\frac{R}{r}\left[H_4 u\left(H_3'' + \frac{1}{r}H_3'\right) + H_3 u\,H_4''\right] = \frac{1}{c^2}H_3 H_4\,\ddot{u}$

Dividing both sides by $H_3 H_4 u$ and multiplying by r/R:

$$\frac{1}{H_3}\left(H_3'' + \frac{1}{r}H_3'\right) + \frac{H_4''}{H_4} = \frac{r}{Rc^2}\frac{\ddot{u}}{u}$$

Because the left side depends on r and z while the right side mixes r and t, consistent separation requires introducing separation constants. We set:

$$\frac{\ddot{u}}{u} = -\omega^2 \quad \Rightarrow \quad \frac{H_4''}{H_4} = -k_z^2 \quad \Rightarrow \quad \frac{1}{H_3}\left(H_3'' + \frac{1}{r}H_3'\right) = -k_r^2$$

with the dispersion-like constraint (from the variable coefficient R/r):

$$k_r^2 + k_z^2 = \frac{\omega^2 R}{r\,c^2}$$

3. Temporal Equation

$$\ddot{u}(t) + \omega^2 u(t) = 0$$

General solution: $\boxed{u(t) = A\cos(\omega t) + B\sin(\omega t)}$

4. Axial Equation

$$H_4''(z) + k_z^2\,H_4(z) = 0$$

General solution: $\boxed{H_4(z) = C\cos(k_z z) + D\sin(k_z z)}$

5. Radial Equation — Bessel Equation

$$H_3'' + \frac{1}{r}H_3' + k_r^2 H_3 = 0$$

This is **Bessel's equation of order zero**: $\frac{d^2H_3}{dr^2} + \frac{1}{r}\frac{dH_3}{dr} + k_r^2 H_3 = 0$

General solution:

$$\boxed{H_3(r) = E\,J_0(k_r r) + F\,Y_0(k_r r)}$$

where J_0 is the Bessel function of the first kind (order 0) and Y_0 is the Bessel function of the second kind (order 0, Neumann function).

For solutions regular at $r = 0$ we set $F = 0$, so:

$$H_3(r) = E\,J_0(k_r r)$$

Key properties used:

$$J_0'(x) = -J_1(x), \qquad Y_0(0) \to -\infty \text{ (singular)}$$

6. General Solution

$$\boxed{f(r,z,t) = \sum_{m,n} J_0\,(k_{r,m}\,r)\left[C_{mn}\cos(k_{z,n}z) + D_{mn}\sin(k_{z,n}z)\right]\left[A_{mn}\cos(\omega_{mn}t) + B_{mn}\sin(\omega_{mn}t)\right]}$$

The variable-coefficient dispersion relation must be satisfied at each r; the dominant mode approximation with $k_r \equiv \alpha/R$ and $k_z \equiv \beta/R$ gives: $\alpha^2 + \beta^2 = \frac{\omega^2 R^2}{c^2}\cdot\frac{R}{r}$

For the **single-mode asymptotic analysis** at $r \sim R$ we treat $R/r \approx 1$ and recover the standard dispersion relation:

$$\omega^2 = c^2(k_r^2 + k_z^2)$$

7. Applying Boundary Conditions Condition 1: $f(2R, 0,0) = 0$

At $t = 0, z = 0$:

$$f(2R,0,0) = J_0(2k_r R)\cdot 1 \cdot A = 0$$

This requires either $A = 0$ (trivial) or $J_0(2k_r R) = 0$.

The zeros of J_0 are $j_{0,n}$: $j_{0,1} \approx 2.4048,\ j_{0,2} \approx 5.5201, \ldots$

So the quantisation condition is:

$$\boxed{k_{r,n} = \frac{j_{0,n}}{2R}}$$

Condition 2: $\partial f/\partial r|_{(0,0,0)} = -R$

$$\frac{\partial f}{\partial r} = -k_r E\, J_1(k_r r)\, H_4(0)\, u(0)$$

At $r = 0$: $J_1(0) = 0$, so this condition is automatically zero for regular Bessel solutions — it constrains the limiting behaviour of the derivative amplitude. Using the small-argument expansion $J_1(k_r r) \approx k_r r/2$:

$$\left.\frac{\partial f}{\partial r}\right|_{r\to 0} = -\frac{k_r^2 E}{2}\cdot r \overset{r\to 0}{\to} 0$$

To satisfy $\partial_r f(0,0,0) = -R$ in the distributional/regularised sense we include the Y_0 term with coefficient F, since $Y_0'(k_r r) = -k_r Y_1(k_r r)$ and $Y_1(x) \sim 2/(\pi x)$ as $x \to 0$:

$$\frac{\partial f}{\partial r}|_{r\to 0} \approx F\cdot(-k_r Y_1(k_r r)) \approx F\cdot\frac{-2}{\pi r}\cdot H_4(0)\cdot u(0)$$

Setting this equal to $-R$ (with $H_4(0) = C = 1$ and $u(0) = A = 1$ normalisation):

$$F\cdot\frac{2}{\pi r}|_{r\to 0^+} = R \;\Rightarrow\; \text{regularised amplitude: } F = \frac{\pi R^2 k_r}{2}$$

For the first mode $k_{r,1} = j_{0,1}/(2R)$:

$$\boxed{F_1 = \frac{\pi R\, j_{0,1}}{4} \approx \frac{\pi \times 3.9228\times 10^{22}\times 2.4048}{4} \approx 7.41\times 10^{22}\ \text{m}}$$

8. Explicit Solution at $t = 0$ With the dominant mode ($n = 1$, $k_z = 0$ for z-independent part):

$$\boxed{f(r,z,0) = \left[E\, J_0\left(\frac{j_{0,1}}{2R}r\right) + F_1\, Y_0\left(\frac{j_{0,1}}{2R}r\right)\right]\cos(k_z z)}$$

The normalisation constant E is set by overall amplitude matching. A convenient choice is $E = R$ so that f carries dimension of length (consistent with magnetic field amplitude interpretation).

9. Construction of $H(r,\theta,z,t)$ Probability Density System

The probability density satisfies:

$$\frac{\partial^2 P(r,z,t)}{\partial t^2} - c^2\frac{\partial^2 P(r,z,t)}{\partial r^2} = 0$$

This is a **1D wave equation in** r for fixed z:

$$P(r,z,t) = P_+(r+ct,z) + P_-(r-ct,z)$$

Boundary condition: $P(2R,0,0) = 0$.

With the squared-amplitude density $P = |\psi|^2$, we identify:

$$P(r,z,t) = |f(r,z,t)|^2$$

At $t = 0$, $z = 0$: $P(r,0,0) = f(r,0,0)^2$, $\quad P(2R,0,0) = f(2R,0,0)^2 = 0$✓

Magnetic Field Amplitude

The full field is reconstructed as (θ-independent):

$$\boxed{H(r,\theta,z,t) = f(r,z,t)\cdot\frac{1}{\sqrt{P(r,z,t)}}\cdot e^{i\phi(r,z,t)}}$$

where ϕ is the phase from the time evolution. In the real formulation:

$$H(r,z,t) = \sqrt{P(r,z,t)} = |f(r,z,t)|$$

The monopole field $\boldsymbol{H}$ in cylindrical coordinates (axial symmetry):

$$\boldsymbol{H}(r,z,t) = H(r,z,t)\,\hat{e}_r = f(r,z,t)\,\hat{e}_r$$

10. Force Calculation at $t = 0$ Definition

From the problem, the force is: $F = \frac{f(r,z,0)}{4\pi r z}$

This has the form of a monopole Coulomb-like force (analogous to $F = q_m B/(4\pi r^2)$ but in cylindrical geometry with $r\cdot z$ replacing r^2).

Evaluation At $t = 0$ with the dominant-mode solution:

$$f(r,z,0) = \left[R\, J_0\left(\frac{j_{0,1}r}{2R}\right) + F_1\, Y_0\left(\frac{j_{0,1}r}{2R}\right)\right]\cos(k_z z)$$

$$\boxed{F(r,z) = \frac{R\, J_0\left(\frac{j_{0,1}r}{2R}\right) + F_1\, Y_0\left(\frac{j_{0,1}r}{2R}\right)}{4\pi r z}\cos(k_z z)}$$

Equality with Magnetic Force (Maxwell)

From Maxwell's equations, the magnetic force per unit volume on a monopole current density $\boldsymbol{K}_m$ is:

$$\boldsymbol{F} = \mu_0\, q_m\, \boldsymbol{H}$$

For the radial component in cylindrical geometry:

$$F_r = \frac{\mu_0\, g^2}{4\pi}\cdot\frac{1}{r^2}$$

where g is the magnetic charge. Setting our expression equal to the Maxwell magnetic force:

$$\frac{f(r,z,0)}{4\pi rz} = \frac{\mu_0 g^2}{4\pi r^2}$$

This gives: $f(r,z,0) = \frac{\mu_0 g^2 z}{r}$

Substituting the solution:

$$\boxed{g^2 = \frac{r}{\mu_0 z}\left[R\,J_0\left(\frac{j_{0,1}r}{2R}\right) + F_1\,Y_0\left(\frac{j_{0,1}r}{2R}\right)\right]\cos(k_z z)}$$

11. Connection to Maxwell's Equations for Monopoles

The extended Maxwell equations including magnetic monopoles are:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho_e}{\varepsilon_0}$$

$$\nabla \cdot \boldsymbol{B} = \mu_0\,\rho_m$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} - \mu_0 \boldsymbol{J}_m$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J}_e + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}$$

In the absence of electric sources ($\rho_e = 0, \boldsymbol{J}_e = 0$), the magnetic field of a monopole satisfies:

$$\Box \boldsymbol{H} = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{H}}{\partial t^2} - \nabla^2 \boldsymbol{H} = 0$$

With the variable-coefficient modification R/r from the monopole geometry, this becomes exactly our governing equation. The wave equation for $\boldsymbol{H}$ in cylindrical coordinates:

$$\frac{R}{r}\left(\frac{\partial^2 H}{\partial r^2} + \frac{1}{r}\frac{\partial H}{\partial r} + \frac{\partial^2 H}{\partial z^2}\right) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$

12. Numerical Values

Quantity	Symbol	Value
Characteristic radius	R	3.9228×10^{22} m
Speed of light	c	2.998×10^{8} m/s
First Bessel zero	$j_{0,1}$	2.4048
Radial wave number	$k_{r,1} = j_{0,1}/(2R)$	3.064×10^{-23} m^{-1}
Characteristic frequency	$\omega_1 = c\,k_{r,1}$	9.19×10^{-15} rad/s
Neumann amplitude	F_1	$\approx 7.41 \times 10^{22}$ m

13. Summary of Solutions

Equation	Solution
$\ddot{u} + \omega^2 u = 0$	$u(t) = A\cos\omega t + B\sin\omega t$
$H_4'' + k_z^2 H_4 = 0$	$H_4(z) = C\cos k_z z + D\sin k_z z$
$H_3'' + \frac{1}{r}H_3' + k_r^2 H_3 = 0$	$H_3(r) = E\,J_0(k_r r) + F\,Y_0(k_r r)$
Full field	$f = H_3(r)H_4(z)u(t)$
Probability density	$P = f^2$, satisfies $\partial_{tt}P = c^2\,\partial_{rr}P$
Force at $t = 0$	$F(r,z) = f(r,z,0)/(4\pi rz)$

14. Asymptotic Behaviour Near $r \to 0$:

$$f(r,z,0) \approx \left[R + F_1\left(\frac{2}{\pi}\ln\frac{k_r r}{2} + \frac{2\gamma}{\pi}\right)\right]\cos(k_z z) + O(r^2 \ln r)$$

where $\gamma = 0.5772\ldots$ is the Euler–Mascheroni constant.

Far field $r \gg R$:

$$f(r,z,0) \approx \sqrt{\frac{2}{\pi k_r r}}\left[E\cos\left(k_r r - \frac{\pi}{4}\right) + F\sin\left(k_r r - \frac{\pi}{4}\right)\right]\cos(k_z z)$$

using the large-argument asymptotics of J_0 and Y_0.

Force at large r:

$$F(r,z) \sim \frac{1}{4\pi rz}\sqrt{\frac{2}{\pi k_r r}}\cos\left(k_r r - \frac{\pi}{4} + \delta\right)\cos(k_z z)$$

which decays as $r^{-3/2}$ — slower than the Coulomb r^{-2} decay, characteristic of cylindrical wave geometry.

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Parameters: $R = 3.9228 \times 10^{22}$ m, $c = 2.998 \times 10^{8}$ m/s